Compounds	Combining elements		Combining weights	
CH_4	C	H	12	4
CO_2	C	O	12	32

Ratio of different mass of hydrogen (4g) and oxygen (32g) that combines with same mass of carbon } 4:32 (or) 1:8——(1)

Now, hydrogen and oxygen combine to form water (H_2O).

Ratio of mass of hydrogen to oxygen = 2:16 (or) 1:8 ———(2)

From 1 and 2, the ratio is the same as that of the first obtained. Thus, the law of reciprocal proportion is illustrated.

11.7.3 Gay Lussac's Law of Combining Volumes

According to Gay Lussac's Law, whenever gases react together, the volumes of the reacting gases bear a simple ratio, and the ratio is extended to the product when the product is also in gaseous state, provided all the volumes are measured under similar conditions of temperature and pressure.

This law may be illustrated by the following example.

It has been experimentally observed that two volumes of hydrogen reacts with one volume of oxygen to form two volumes of water as shown in the figure 11.12.

The ratio of volume which gases bears is **2:1:2** which is a simple whole number ratio.

Activity 6

Nitrogen combines with hydrogen to form ammonia (NH_3). Illustrate Gay Lussac's law using this example.

11.8 Quantum Numbers

When you specify the location of a building, you usually list which country it is in, which state and city it is in that country.

Just like we have four ways of defining the location of a building (country, state, city, and street address), we have four ways of defining the properties of an electron, i.e.four quantum numbers.

Thus, the numbers which designate and distinguish various atomic orbitals and electrons present in an atom are called quantum numbers.

Four types of Quantum number are as follows:

Quantum Number	Symbol	Information conveyed
Principal quantum number	*n*	Main energy level
Azimuthal quantum number	*l*	Sub shell/ shape of orbital
Magnetic quantum number	*m*	Orientation of orbitals
Spin quantum number	*s*	Spin of the electron

You will learn more details about this in higher classes.

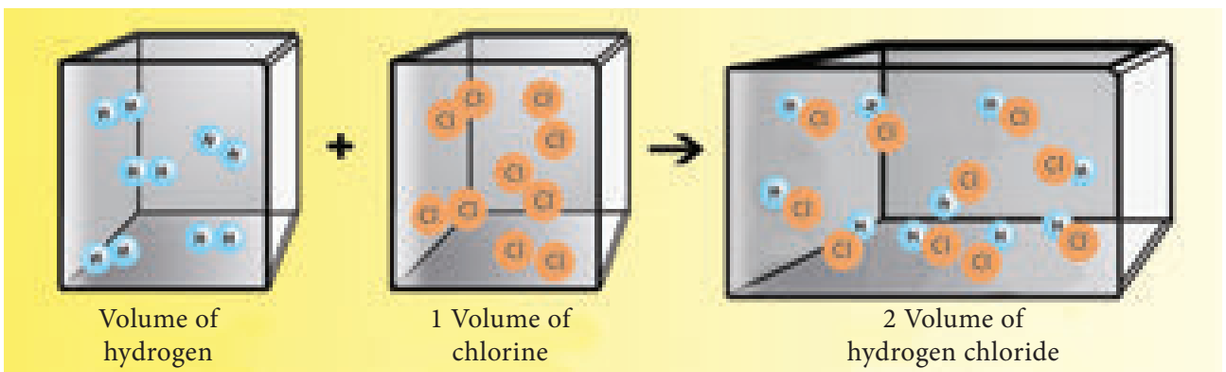


Figure 11.12 One volume of hydrogen react with one volume of chlorine to give two volumes of hydrogen chloride.

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